PEFN: A Patches Enhancement and Hierarchical Fusion Network for Robust Vehicle Reidentification

Wenying He, Feiyu Wang, Yude Bai, Neal N. Xiong, *Senior Member, IEEE*, Guangquan Xu, *Member, IEEE*, and Fei Guo

***Abstract*—Vehicle reidentification (Re-ID), which is a significant application in the Internet of Things, aims to accurately retrieve the remaining images of a given vehicle across different cameras views. The improvement in vehicle Re-ID performance largely stems from better addressing the issues of interclass similarity and intraclass variance. Existing methods, relying solely on max or average pooling after using attention modules, fail to obtain significantly complete and pure global and local features, and neglect the false guidance that some unique individual information on images bring to Re-ID. Moreover, models combining global and local features have shown good results in vehicle Re-ID, but these successes neglect the interaction between features across different convolutional layers, resulting in the loss of crucial details for vehicle Re-ID. To tackle these issues, we introduce a patches enhancement and hierarchical fusion network (PEFN) based on a multibranch architecture, divided into a global and local attention supplement (GLAS) branch, and an enhanced hierarchical feature fusion (EnHi) branch. The GLAS branch, through the identity-related feature remodeling (IDFR) module's staged supplementation of spatial and channel features, has achieved the enhancement of both global and local features and effectively mitigated the negative impacts of individual information. The EnHi branch enhances the robustness of feature representation by interacting hierarchical features. Extensive experiments on two large-scale vehicle Re-ID datasets demonstrate that our PEFN method outperforms state-of-the-art vehicle Re-ID approaches. Specifically, without utilizing extra data and reranking, our model achieves 85.15% mean average precision on the VeRi776 dataset. Code is available at https://github.com/711L/PEFN.**

***Index Terms*—Feature fusion, Internet of Things, unique individual information, vehicle reidentification (Re-ID).**

I. Introduction

GIVEN the query image, the task of vehicle reidentification (Re-ID) [1], [2], [3], [4], [5], [6], [7], [8] is to find the remaining images of same identity captured by different cameras from a large gallery set. It plays an important role in the construction of intelligent transportation systems, which is a hot field in the Internet of Things. Currently, with the development of the economy and the popularity of electric vehicles, the demand for vehicle management and Re-ID in public security systems has been increasing day by day. The introduction of large-scale datasets [9], [10], [11], [12], [13] and the rise of deep learning have accelerated progress in research on vehicle Re-ID. However, due to their unique interclass similarity and intraclass variance problems, the performance of vehicle Re-ID falls far short of person Re-ID [14], [15], [16]. This indicates that there are still significant challenges and research prospects in vehicle Re-ID.

The human process of distinguishing objects first involves initial categorization based on prominent features, followed by analysis of finer details, context integration, memory matching, and cognitive processing. This process can be summarized as "prominent first, then fine-grained." Most existing vehicle Re-ID methods adhere to this principle, employing a dual-branch architecture to address the aforementioned two issues. In such architecture, one branch is often used to extract global features, while the other branch rigidly partitions the feature map, utilizes an additional network to locate key-points, or constructs nodes on the feature map to form a graph to extract fine-grained discriminative features. This forms a Re-ID pattern where discriminative regional features combine global features.

However, these methods do not eliminate the negative impact of unique individual information on the feature extraction process. As shown in Fig. 1(a), the positive samples of the anchor image contain grass and streetlights, which are irrelevant information regarding the identity of the vehicle. The negative samples contain individual information, such as streetlight and fence in Fig. 1(b). The unique individual information may generate spurious intraclass and interclass

Received 10 February 2025; revised 25 March 2025; accepted 10 April 2025. Date of publication 15 April 2025; date of current version 9 July 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFB2703800; in part by the National Natural Science Foundation of China under Grant 62302148, Grant 62172297, and Grant U22B2027; in part by the Hebei Natural Science Foundation under Grant F2024202076; in part by the Hebei Yanzhao Huang Jin Tai Talents Program (Postdoctoral Platform) under Grant B2024003002; in part by the Hainan Province Science and Technology Special Fund under Grant ZDYF2024GXJS008; in part by the Guangxi Science and Technology Plan Project (Guangxi Science and Technology Base and Talent Special Project) under Grant AD23026096; and in part by the Supplementary Funds for Performance of Hebei Province Key Laboratory of Big Data Calculation under Grant 22567603H. *(Corresponding authors: Guangquan Xu; Fei Guo.)*

Wenying He is with the School of Artificial Intelligence, Hebei University of Technology, Tianjin 300401, China, and also with the Hebei Province Key Laboratory of Big Data Calculation, Hebei University of Technology, Tianjin 300130, China (e-mail: hwying1234@hebut.edu.cn).

Feiyu Wang is with the School of Artificial Intelligence, Hebei University of Technology, Tianjin 300401, China (e-mail: 202332805032@stu.hebut.edu.cn).

Yude Bai is with the School of Software, Tiangong University, Tianjin 300387, China (e-mail: baiyude@tiangong.edu.cn).

Neal N. Xiong is with the Department of Mathematics and Computer Science, Northeastern State University, Tahlequah, OK 74464 USA (e-mail: xiongnaixue@gmail.com).

Guangquan Xu is with the Tianjin Key Laboratory of Advanced Networking (TANK), Tianjin University, Tianjin 300350, China (e-mail: losin@tju.edu.cn).

Fei Guo is with the School of Computer Science and Engineering, Central South University, Changsha 410083, China (e-mail: guofei@csu.edu.cn).

Digital Object Identifier 10.1109/JIOT.2025.3561186

2327-4662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

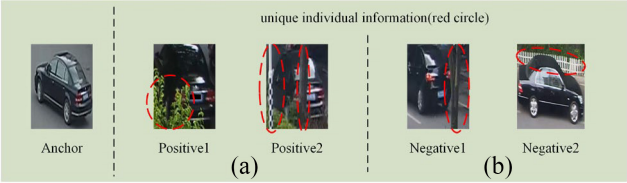


Fig. 1. Unique individual information is contained in the vehicle images. (a) Positive samples of the anchor image. (b) Negative samples of the anchor image. The red circle represents the unique individual information of the image.

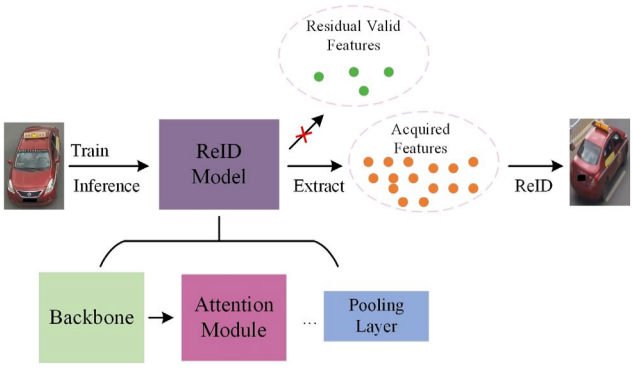


Fig. 2. Completeness issues in feature extraction using attention modules and pooling operations.

relationships during the feature exploration, which leads to inefficient Re-ID performance. Furthermore, relying solely on pooling after using attention modules leads to an incomplete feature extraction method. As shown in Fig. 2, the model only uses the features after pooling for Re-ID, but loses the residual valid features. Due to the incompleteness of the extraction method, the global and local branch lacks pathways for attention supplementation, leading to incomplete features, which in turn hinders the improvement of Re-ID performance. Moreover, the local branch and the global branch only converge at the end, without utilizing contextual information from different layers. Thus, the general implementation of this vehicle Re-ID pattern lacks consideration for the completeness and validity of both global and local features, as well as the interaction of contextual information.

Toward these issues, in this article, we propose a multi-branch model named patches enhancement and hierarchical fusion network (PEFN). First, original vehicle images pass through the global and local attention supplement (GLAS) branch, undergoing the backbone network followed by the identity-related feature remodeling (IDFR) module. The IDFR module calculates channel and spatial attention (SA) for small patches and enhances the feature maps at each stage based on residual augmentation. It supports procedural attention supplementation pathways to retain complete features and helps the model reduce the negative impact of unique individual information in the image, thereby enhancing the completeness and purity of the features. Following the idea of bilinear pooling, the enhanced hierarchical feature fusion (EnHi) branch integrates features obtained from different layers of the network, thereby promoting the interaction of contextual information to improve feature representation.

To validate the proposed PEFN model, experiments are conducted on two datasets, VeRi776 and VehicleID. The former consists of 776 vehicles from 20 cameras. The latter includes 113 346 images from 13 164 vehicles, which are split into small, medium, and large subsets. The results demonstrate that PEFN achieves 85.15% mean average precision (mAP), 98.03% CMC1, and 98.87% CMC5 on the VeRi776 dataset. It outperforms all compared methods, such as random access memory (RAM), PVEN, and SAN, particularly in terms of mAP, CMC1, and CMC5 scores. Furthermore, PEFN significantly improves performance through feature map patching, showing strong retrieval capabilities even with occlusions and similar vehicle shapes. Ablation studies reveal that different components of PEFN, such as the IDFR module, attention mechanisms, and feature map patching, contribute to performance improvement (about the increase of 0.57%–2.02% in mAP).

In a nutshell, the contributions of our work can be summarized as follows.

1) We propose a PEFN to capture both global and local features, as well as contextual information for vehicle Re-ID.
2) We have designed the IDFR module to compute attention based on patches and recycle residual attention, eliminating the negative effects of unique individual information and enhancing the completeness of features.
3) Experiment results on two large-scale vehicle datasets show that the proposed method is superior to state-of-the-art vehicle Re-ID approaches.

The remainder of this article is organized as follows. Section II introduces the related work. Section III elaborates our proposed method. Section IV presents the experimental results to show our method's superiority. Section V concludes this article.

II. Related Work

A. Global Feature Learning Methods

The initial implementation of vehicle Re-ID mainly relied on extracting effective global features. Early methods [13], [17], [18], [19], [20], [21] applied convolutional neural networks (CNNs) to learn the vehicle's global features. These methods often included a backbone network (such as GoogleNet [22] and VGGNet [23]) and a pooling layer, which aggregates information from the entire image, providing a compact representation of the vehicle's overall appearance. The role of global features has proven effective in vehicle Re-ID, significantly promoting subsequent research. However, the global pooling mechanism tends to suppress fine-grained details and hinders the discovery of the most distinctive local features, which severely impacts model performance in certain cases. This limitation is particularly noticeable in visually similar vehicles, where subtle local features are critical for distinguishing between them.

Network Architecture Enhancements: To address the limitations of global pooling, several studies have proposed novel architectures designed to better capture and utilize global features. Zhu et al. [17] introduced the short and

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

densely connected convolutional network (SDC-CNN) for vehicle Re-ID. This architecture features short and dense units (SDUs) that enhance feature learning by incorporating multiple convolutional layers with limited connections, enabling the network to learn more discriminative features with fewer parameters. Yan et al. [13] proposed an improved framework with multigrain ranking constraints, introducing generalized pairwise ranking and multigrain list ranking. This method outperforms traditional approaches by considering multiple relational constraints between images, improving vehicle matching accuracy.

Loss Function Innovations: Innovations in loss functions have also significantly contributed to the improvement of global feature extraction. Zhang et al. [20] enhanced traditional triplet-wise training by incorporating a classification-oriented loss, improving the model's ability to capture the relationships between positive and negative samples more effectively. Their approach, along with a new triplet sampling method, has demonstrated substantial performance improvements on vehicle Re-ID benchmarks. Kumar et al. [21] tackled viewpoint variations by using triplet embeddings to match vehicles across different camera views. Their method, which uses identity-level annotations, is both computationally efficient and effective.

Dataset Utilization and Progressive Learning: The creation and use of large-scale datasets have been pivotal in improving the effectiveness of vehicle Re-ID models. Zheng et al. [18] proposed the VehicleNet dataset, which is constructed from multiple existing datasets to overcome intraclass variation across camera views. They introduced a progressive learning strategy that first learns a generic feature representation across datasets and then fine-tunes the model on the target dataset, minimizing distribution discrepancies. This approach led to state-of-the-art performance in the AICity Challenge.

In summary, early vehicle Re-ID methods focused primarily on global feature extraction using CNNs, demonstrating the effectiveness of global features in vehicle matching tasks. However, the limitations of global pooling, such as its inability to capture fine-grained details, have led to the development of innovative solutions in network architecture, loss functions, and dataset utilization. The SDC-CNN architecture [17], multigrain ranking constraints [13], enhanced triplet loss functions [20], [21], and large-scale datasets like VehicleNet [18] have significantly advanced the field. Despite these advances, challenges remain, particularly in capturing local features and handling environmental variations, such as illumination and viewpoint changes. Researches should focus on integrating local and global feature extraction techniques to address these challenges, improving both the accuracy and robustness of vehicle Re-ID systems.

B. Local Feature Learning Methods

Therefore, current researches on vehicle Re-ID primarily achieve the combination of global and local features through multibranch structures. These approaches are divided into different categories based on their treatment of local features. Despite their significant contributions, these methods still face challenges, such as the lack of robust integration between global and local features, high computational cost, and difficulties in capturing contextual information across different network layers. Additionally, most approaches heavily rely on annotated data or complex auxiliary modules, which limit their scalability and practicality in real-world scenarios.

Part-Based Methods: References [24], [25], [26], [27], [28], [29], [30], [31], and [32] capture specific regional parts from images to extract discriminative local features. For example, PVEN [24] and GLNet [25] utilize additional segmentation networks to divide vehicles into several visible specific parts, enabling feature comparison of common visible areas even for images from different viewpoints. While effective, these methods heavily depend on precise segmentation and viewpoint alignment, which may not generalize well to unconstrained environments. Khorramshahi et al. [26] proposed a combination of viewpoint prediction and key-point detection modules to locate local features, along with an adaptive key-point selection module to learn relationships between parts. This approach significantly improves accuracy but introduces high complexity due to its reliance on auxiliary modules and large amounts of annotated training data. Stripe-based approaches, such as [29], divide feature maps horizontally into several meaningful parts to capture local features. Similarly, Liu et al. [30] uniformly divided the feature map into three regions along the vertical axis, assuming that most vehicle images are well-aligned. These methods achieve good performance by pooling features from predefined regions but fail to account for irregularities in vehicle orientation or occlusion. Regardless of the specific partitioning approach, the learning of local features in these methods tends to be fragmented, limiting their ability to capture coherent structural information.

Graph-Based Structural Feature Methods: These methods integrate graph networks with CNNs to capture the interplay between different vehicle parts. By leveraging graph networks, they establish nodes on segmented regions, creating a structured representation that captures relationships between parts. For instance, the global structural graph [33] splits feature maps generated by CNNs, constructs an edge-adaptive global structure graph, and utilizes graph convolutional networks for relational reasoning, forming fine-grained representations of vehicles. LABNet [34] introduces a local graph aggregation module after feature extraction, treating spatial regions in the feature map as nodes and constructing local neighborhood graphs for feature aggregation. This approach effectively captures localized structural details, yet it struggles to balance local and global feature integration. GSAN [35] enhances performance by capturing structural attribute features and exploring relationships between parts and attributes, while HPGN [36] stacks a series of spatial graphs to explore the spatial significance of feature maps at multiple scales. These methods excel in modeling the relationships between parts but often overlook the critical interaction between global and local features. Addressing this gap, GiT [37] couples graph networks with Transformer [38], enabling global and local features to guide each other, thus improving contextual understanding. However, the added complexity of

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

Transformer-based architectures introduces significant computational overhead, which may hinder real-time applications.

Self-Supervised Feature Extraction Methods: These methods enhance feature representation by generating supervision signals directly from input data, offering an unsupervised approach to vehicle Re-ID. For example, SAVER [39] employs a variational autoencoder to coarsely reconstruct images, subsequently performing a weighted sum of the original image and the reconstructed details to enhance attention on discriminative features. While effective in certain scenarios, its reliance on reconstruction quality may limit its robustness. Li et al. [40] introduced an interpretable attention module to facilitate the discovery of geometric features, thereby improving vehicle instance differentiation. However, this approach's heavy focus on geometric attributes may overlook other crucial contextual cues. SSBVER [41] leverages a self-supervised student–teacher network pair and knowledge distillation to improve accuracy and scalability, demonstrating potential in reducing reliance on annotated data. Nevertheless, the computational cost associated with self-supervised learning frameworks, particularly when dealing with extremely large datasets, poses a significant challenge.

In tasks involving vehicle Re-ID, the computation of self-attention mechanisms incurs substantial time costs [42], [43], making CNN-based approaches more practical in achieving efficient performance. Considering that CNNs still have considerable room for improvement, we designed a multibranch network based on CNNs and demonstrated its effectiveness in this article. Specifically, we recycle small-patch features to enhance both global and local features, allowing them to mutually reinforce each other. Our approach not only focuses on the connections between small blocks but also considers the contextual information between different network layers. This design enables us to fully leverage the hierarchical structure of CNNs, improving feature representation in a manner understandable to humans. Most existing methods focus on increasing model complexity to enhance the diversity of discriminative features, yet they fail to consider that the relevance of features to the Re-ID task varies. In other words, while our method enhances feature diversity through a multibranch structure, it also remodels these features via the IDFR module, thereby decoupling the important identity-related discriminative features from the rest. This is the most significant distinction of our method from others. As a result, compared with other models, our method achieves significant improvements in performance and accuracy, demonstrating its potential as a robust and efficient solution for vehicle Re-ID.

III. Our Proposed PEFN

As shown in Fig. 3, our model includes a backbone network, a GLAS branch, and an EnHi branch. We select ResNet50 [44], celebrated for its robust performance and distinctive residual connections, as the backbone, pretrained on ImageNet [45]. This architecture is divided into two main branches: the GLAS branch processes feature maps from the backbone network, while the EnHi branch manages outputs from the first three residual blocks in the backbone's layer4. Notably, both branches are trained using a combined approach of cross-entropy loss [46] and batch-hard triplet loss [47].

A. GLAS Branch: Global and Local Attention Supplement Branch

This branch is designed for extracting more comprehensive global and local features for training and inference. We introduce the IDFR module, which plays a central role in GLAS branch. In Fig. 4, suppose the input to the IDFR module is a feature map $F \in \mathbb{R}^{C\times H\times W}$, where $\mathcal{C}$, $\mathcal{H}$, and $\mathcal{W}$ refer to the number of channels, height, and width, respectively. The feature map F is sequentially divided into n patches along the height and width dimensions. The size of the height and width of each patch is $(H/p, W/p)$, where $p = n^{1/2}$. Then, the set of patch feature maps $F_n = \{f_1, f_2, \ldots, f_i, \ldots, f_n\}$ are obtained. There is no doubt that single-stage attention module plays a role in extracting the significance of feature maps. However, due to the computation conducted in a manner predetermined by humans, the optimality of the model's focus outcomes can be relative in many scenarios. For example, f_a and f_b are two important features, where f_a aligns with the human-defined optimum, leading to f_b being discarded. Therefore, this strategy can impact the efficacy of the model. To avoid the circumstance, it is necessary to recycle residual attention to enhance the features. Staged extraction of channel attention (CA) and SA is more suitable for recycling residual attention, thereby obtaining complete and significant features. The PEFN adopts the simple and effective CBAM [48] as its attention mechanism. For each $f_i \in F_n$, sequentially feed it into the CA module followed by the SA module, thereby obtaining feature maps with differential attention across various regions and channels. The outputs from these two modules are denoted as f_i' and f_i'', respectively. The feature pairs that conform to the feature enhancement pairing are (f_i, f_i''), (f_i, f_i'), and (f_i', f_i''). For each pair, ro_i, r_{i1}, and r_{i2} are obtained by subtracting the latter from the former. Therefore, the differences of all patches are aggregated according to (1), as follows:

$$\begin{aligned} r_o &= \mathrm{Agg}(ro_1, ro_2, \ldots, ro_n) \\ r_1 &= \mathrm{Agg}(r_{11}, r_{21}, \ldots, r_{n1}) \\ r_2 &= \mathrm{Agg}(r_{12}, r_{22}, \ldots, r_{n2}). \end{aligned} \tag{1}$$

According to (2), the original outputs of CBAM are aggregated as follows:

$$\mathrm{mb} = \mathrm{Agg}\left(f_1'', f_2'', \ldots, f_n''\right) \tag{2}$$

where mb represents an incomplete representation of critical feature map, potentially lacking essential information in either the channel dimension, the spatial dimension, or both. According to (3), the three types of residual features are fused, as follows:

$$\mathrm{rub} = r_1 \otimes r_2^T + r_o \tag{3}$$

where r_1 denotes the supplementary features in the channel dimension, r_2 signifies the supplementary features relevant to the spatial dimension, and r_o represents the supplementary features in the combined channel–spatial dimensions.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

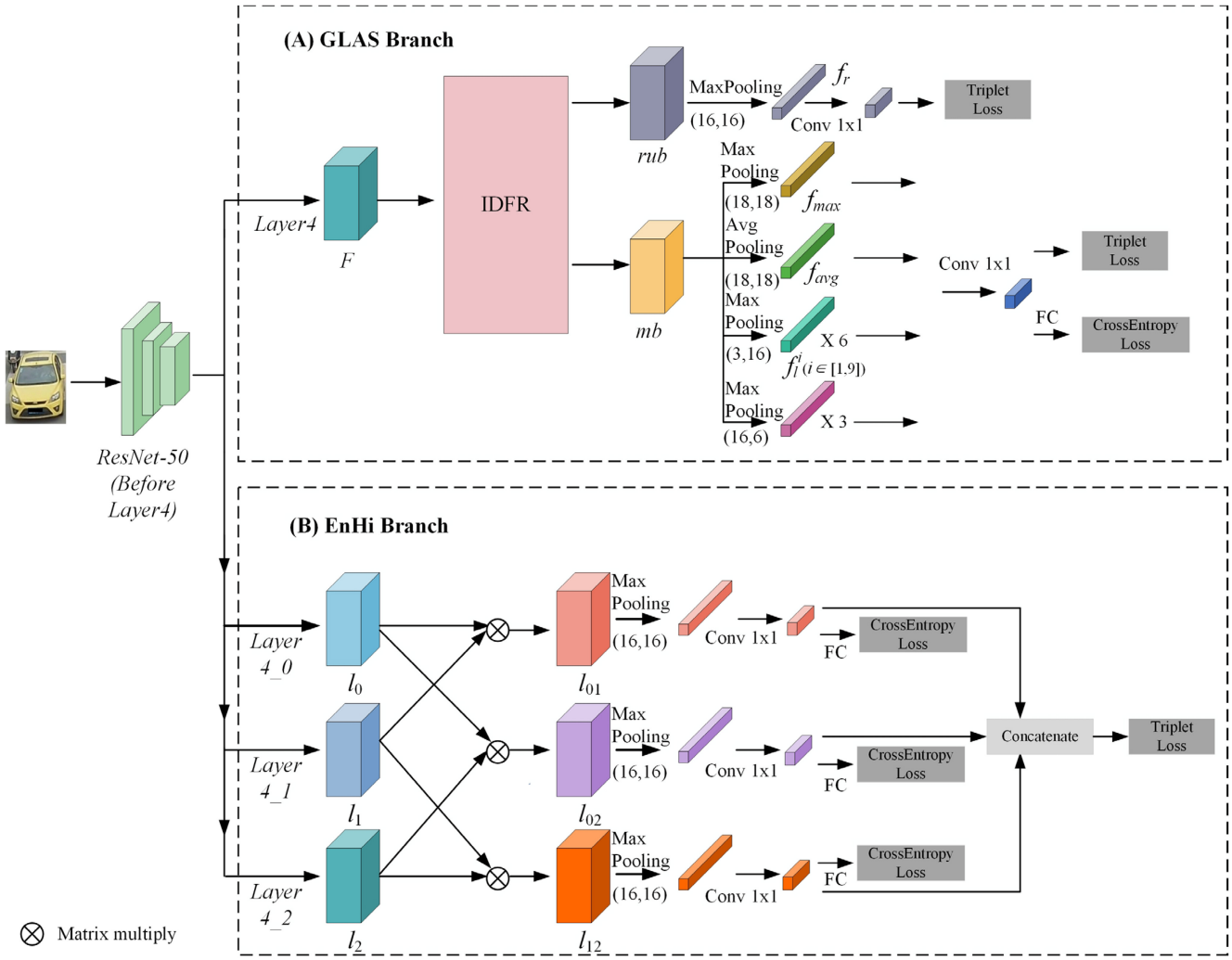


Fig. 3. Framework of the proposed PEFN. The network employs ResNet-50 as backbone network, and this architecture is divided into two branches, namely, GLAS branch and EnHi branch. (a) GLAS branch effectively extracts enhanced global features and local features. (b) EnHi branch is responsible for fusing cross-level features to obtain a rich contextual representation. The Conv and FC represent a convolutional layer and a fully connected layer, respectively.

The size of mb and rub is maintained in parity with the feature maps originally fed into the IDFR module. According to (4), two aggregated methods are applied to mb to form the global features. Residual valid features are obtained by

$$f_{\max} = \text{Maxpool}(\text{mb}), f_{\text{avg}} = \text{Avgpool}(\text{mb}) \tag{4}$$

$$f_r = \text{Maxpool}(\text{rub}). \tag{5}$$

$f_{\max}$ and f_{avg} denotes the significant global feature and the robust local feature. f_r represents the residual features utilized by the model. To enable the model to learn more robust and discriminative feature representations, two pooling windows are employed to perform pooling on mb. Following a comprehensive analysis of the vehicle image structure and extensive experimentation, the feature maps post-pooling are rigidly divided into six horizontal partitions and three vertical partitions. The local features are constructed according to (6), as follows:

$$f_l^i = \text{Maxpool}\big(\text{partition}_i\big), i \in [1, \ldots, 9]. \tag{6}$$

The pseudocode for the IDFR module is shown in Algorithm 1 as follows.

Algorithm 1 IDFR Module

Input: $F \in \mathbb{R}^{C\times H\times W}$

1: F is divided into n parts along the H and W dimensions to obtain the set F_n in the Partition phase.

2: In the Enhancement phase, **for** $f_i \in F_n$ **do**

3: Extraction of features from *CA* and *SA* module:

$f_i' = CA(f_i), f_i'' = SA(f_i')$.

4: Constructing the difference of related feature pairs:

$r_{i1} = f_i - f_i'$, $r_{i2} = f_i' - f_i''$, $ro_i = f_i - f_i''$.

5: In the Aggregation phase:

Obtain the corresponding aggregations of r_o, r_1 and r_2 by Eq. (1).

Obtain the main feature mb by Eq. (2).

Calculate the fused enhanced feature *rub* by Eq. (3).

Output: $mb \in \mathbb{R}^{C\times H\times W}$, $rub \in \mathbb{R}^{C\times H\times W}$

B. EnHi Branch: Enhanced Hierarchical Feature Fusion Branch

Most vehicle Re-ID efforts do not effectively utilize the abundant contextual information. Multiscale feature fusion is a widely used method for feature integration in computer vision tasks [49]. It extracts features from layers of varying depths

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

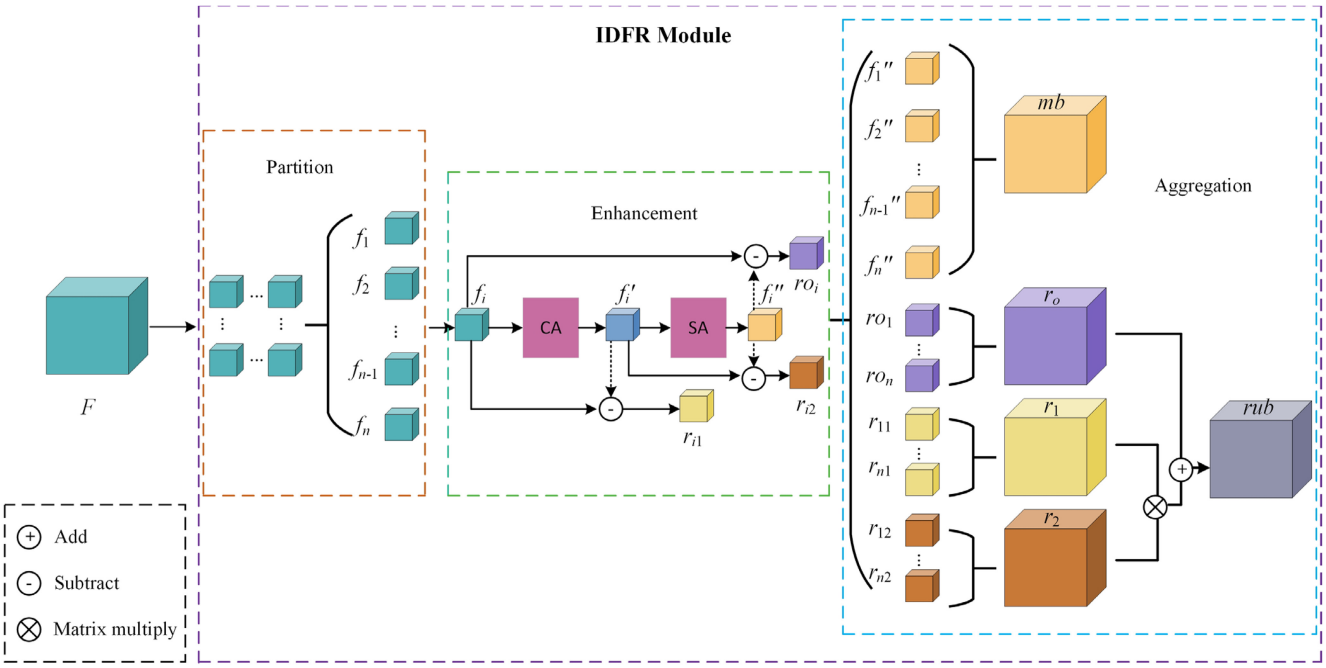


Fig. 4. Overview of the IDFR module. It comprises three stages: partition, enhancement, and aggregation the feature maps. The enhancement phase incorporates CA and SA mechanisms to refine the feature representation.

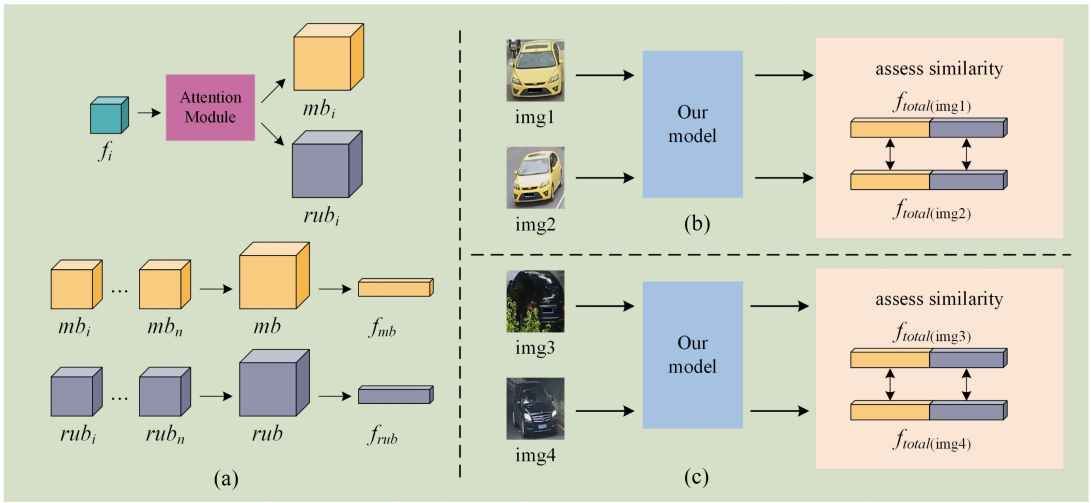


Fig. 5. Illustration of the feature processing workflow and mechanism of the IDFR Module. (a) Each patch extracts features and is then aggregated into mb and rub, which are subsequently constructed into f_{mb} and f_{rub}. (b) Model extracts enhanced image features for similarity assessment. (c) Model alleviates the negative effects of unique individual information to assess similarity and enhance Re-ID performance.

within the network and fuses them through concatenation or weighted summation to capture contextual information. This approach is exemplified in models like U-Net [50] for semantic segmentation and YOLOV5 for object detection. The bilinear pooling [51] fuses features from different scales, providing a richer feature expression capability than traditional max pooling and average pooling. It is very suitable for capturing fine-grained features in vehicle Re-ID and can effectively utilize contextual information between different layers. Bilinear pooling typically requires two feature extractors, which undoubtedly increases the burden on the model. To address this issue, as shown in Fig. 3, EnHi branch adopts a homologous bilinear pooling approach, integrating the outputs from the first three bottleneck blocks of layer4 in the ResNet backbone network

$$l_{01} = l_0 \otimes l_1, l_{02} = l_0 \otimes l_2, l_{12} = l_1 \otimes l_2 \tag{7}$$

where l_0, l_1, and l_2 represent the outputs of the first three bottleneck blocks. l_{01}, l_{02}, and l_{12} represent the results of their pairwise interactions.

C. Feature Enhancement and Reduction of Unique Individual Information

As shown in Fig. 5(a), in our model, after the feature maps are split, each patch is input into the attention module to obtain

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

the corresponding mb_i and rub_i. Then, they are aggregated into mb and rub. The following discussion will ignore the subsequent operations on mb and rub as shown in Fig. 3, and will use f_{mb} and f_{rub} to represent all the results of different pooling operations instead. f_{mb} and f_{rub} are concatenated to form f_{total} for assessing the similarity between two images. The similarity assessment between image i and image j is formulated as follows:

$$f_{\text{total}} = \text{Concat}(f_{\text{mb}}, f_{\text{rub}}) \tag{8}$$

$$\text{similarity} = D\left(f_{\text{total}(i)}, f_{\text{total}(j)}\right) \tag{9}$$

where D represents the distance metric function, and $f_{\text{total}(i)}$ represents the overall features of the image i.

When disregarding the individual information on the images, as shown in Fig. 5(b), img1 and img2 are used to determine whether they are images of the same vehicle. f_{mb} denotes the significant global feature and the robust local feature. f_{rub} represents the supplement of information in both the spatial and channel dimensions, facilitating the Re-ID process of two images.

As shown in Fig. 5(c), img3 contains unique individual information that affects identity determination. We assume that the features containing unique individual information are divided into a patch and marked as f_i. When f_i flows through the augment process as shown in Fig. 4, due to the presence of an attention module, the mb_i contains salient features with unique individual information, which are meaningless for similarity comparison. On the contrary, rub_i is constructed from the differences at various stages, which effectively mitigates the impact of unique individual information while also supplementing information in both the channel and spatial dimensions. When constructing mb and rub and pooling them to form f_{total}, there will be a significant difference in f_{mb} between img3 and img4. According to (8) and (9), their f_{rub} enhances the Re-ID performance by mitigating the impact of unique individual information.

D. Loss Function

Vehicle Re-ID is not regarded as a simple classification task or a metric learning, but should combine both to enhance the model's capability to learn discriminative features. In the dual branches mentioned above, the features are processed through pooling, convolutional dimensionality reduction, and batch normalization before the applying the triplet loss. Then, the features are passed through a fully connected layer and fed into the cross-entropy loss with label smoothing. Therefore, the proposed PEFN's total loss function is formulated as follows:

$$L_{\text{total}} = \alpha L_{\text{CE}} + \beta L_{\text{BHT}} \tag{10}$$

where L_{CE} and L_{Triplet}, respectively, denotes cross-entropy loss and batch-hard triplet loss. α and β are manually setting constants used to keep balance of two loss functions.

The cross-entropy loss function [46] is formulated as follows:

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{K} y_{ij} \log \frac{e^{W_j^T X_i}}{\sum_{k=1}^{K} e^{W_k^T X_i}} \tag{11}$$

where X is the training samples set and X_i is the ith training sample; N is the number of the training samples and K is the number of the vehicle classes; y_{ij} denotes the true label information of sample X_i for the jth class; and W_j is the weight of the jth classifier.

The triplet loss function aims to reduce the distance between samples of the same classes and enlarge the distance between samples of different classes. The initial implementation of triplet loss uses excessive numbers of triplets, most of which do not contribute significantly to improving the model. To solve this issue and further improve the ability of triplet loss to distinguish difficult samples, Hermans et al. [47] proposed batch-hard triplet loss, which aims to compute by selecting the hardest positive and the hardest negative samples within each mini-batch. The batch-hard triplet loss can be formulated as follows:

$$L_{\text{BHT}} = \sum_{i=1}^{V}\sum_{a=1}^{K}\Bigg[\max_{p=1\ldots k_{p\neq a}} D\left(X_i^a, X_i^p\right) - \min_{\substack{j=1\ldots p \\ j\neq i}}{}^{n=1\ldots k} D\left(X_i^a, X_j^n\right) + m\Bigg]_+ \tag{12}$$

where $[\cdot]_+ = \max(0,\cdot)$ is a hinge function; V is the number of vehicle identities and K is the number of images for each identity in a mini-batch; $D(a, b) = \|a - b\|_2$ is the Euclidean distance; X^a, X^p, and X^n represent, respectively, the extracted embedding of anchor, the hardest positive, and the hardest negative samples. Our model adopts batch-hard triplet loss as triplet loss.

IV. Performance Analysis

To validate the superiority of the proposed method, It is compared with state-of-the-art vehicle Re-ID methods on two widely used datasets, namely, VeRi776 [9], VehicleID [10]. We report our findings in terms of percentage of mAP [52] and cumulative match curve (CMC) at rank 1 and 5 to assess the accuracy performance.

mAP: We first compute the average precision (AP) for each query: $\text{AP} = (1/N)\sum_{k=1}^{K} P(k)\cdot \text{rel}(k)$. N is the number of correct matches. K is the total number of retrieval results. $P(k)$ is the precision at cut-off point k. If the kth retrieval result is a correct match, $\text{rel}(k)$ equals 1. The mAP is the average AP over M queries: $\text{mAP} = [1/M]\sum_{i=1}^{M} AP_i$.

Cumulative Matching Characteristic: The CMC curve measures the probability that the correct match is included in the top-k retrieval results. It can be formulated as: $\text{CMC@}k = (1/Q)\sum_{j=1}^{Q} s(k)$. Q is the total number of query images. If the correct match appears in the top-k retrieval results, the indicator function $s(k)$ equals 1, otherwise it is equal to 0.

A. Datasets

The datasets utilized for validation are VeRi776 and VehicleID.VeRi776 comprises images of 776 vehicles captured under 20 different cameras. The dataset is split into 37 778 training images, 1678 query images, and 11 579 gallery images. The VehicleID dataset is another widely used dataset in the eld of vehicle Re-ID. The training set of this dataset

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

TABLE I
VALIDATION OF THE EFFECTIVENESS (%) OF FEATURE MAP PATCHING ON VERI776

isPatching	mAP	CMC1	CMC5
PF	83.44	96.78	98.45
PT	**85.15**	**98.03**	**98.87**

consists of 113 346 images of 13 164 subjects. The original test set is divided into three subsets: small, medium, and large, which contains 800, 1600, and 2400 subjects, respectively. Moreover, no specific method for dividing the three subsets into query sets and gallery sets; thus, randomly select one image for each subject to form query sets and all remaining images of this subject are used to construct gallery sets.

B. Implementation Details

For the baseline network, we use the ResNet-50 model parameters pretrained on ImageNet to initialize our network. In layer4, cancel the original downsampling operation and split layer4 into three parts to be utilized in EnHi branch. All input images are resized to 288 × 288, and then data augmentation methods are adopted, including random horizontal flipping, normalization, random erasing in training stage, and normalization in inference stage. To construct training batches, we set $V = 7$ and $k = 9$ on two datasets, resulting in the batch size of 63. In the training phase, Adam optimizer is used to optimize our model and the weight decays are set to 5×10^{-4}. There are 700 epochs for the training process. The initial learning rate is 2.6×10^{-4} and the learning rate is scheduled to decay at epochs 240, 400, 540, and 600 by a factor of 0.1 using the MultiStepLR scheduler, aiming to balance rapid initial convergence with fine-tuning capabilities throughout the training process. However, to prevent overfitting and ensure optimal model performance, we employed the early stopping method. Training was halted when the model's performance on the validation set ceased to improve. In practice, this stopping criterion was triggered around epoch 660. The optimal parameter of partitions for IDFR module's feature map segmentation is set as $N = 4$ based on the experiment results. Our model's training time on the VeRI776 dataset and the VehicleID dataset is 8 and 33 h, respectively. Our work is implemented on one NVIDIA RTX 4090 with 24-GB RAM.

C. Effectiveness of Feature Map Patching

As shown in Fig. 3, the feature map underwent patching operations in the IDFR module, for which we conducted experiments to verify the effectiveness of the patching operation. The results of the verification on VeRi776 are shown in Table I. PF and PT represent PEFN model without and with patching operation, respectively, both of which have an attention enhancement step. The results show that the patching operation greatly improves the model's performance. The reason is that the pooling results of large-scale feature maps may not necessarily reflect the universality of the feature map, whereas after the patching operation, the smaller multiple feature maps can better find the universality that fits each block during the pooling operation.

D. Comparison With State-of-the-Art Methods

As vehicle Re-ID research methods advance, the combination of global and local features has become a prevalent approach. Therefore, we primarily selected vehicle Re-ID methods based on multibranch global and local features for comparison. They can be categorized into the following classes.

1) *Part-Based Methods:* PVEN [24] uses U-Net parses a vehicle image into four views to learn subtle local information. AAVER [26] captures discriminative local information by key-point detection and orientation estimation. SAN [29] utilizes rigid partitioning to learn part-level features. RAM [30] learns regional features from three overlapping local regions by horizontally dividing the feature map. HCI-Net [53] rigidly partitions the feature maps and utilizes the designed channel and spatial heterogeneous context interaction modules to obtain global and local contexts, thereby enhancing the effect of attention learning.
2) *Graph-Based Structural Feature Methods:* HPGN [36] combines multiple SGN through a pyramid structure to discover the importance of feature maps at different scales. GiT [37] learns global and local features through Transformer layers and the LCG module. PCRNet [54] segments vehicle parts, extracts regional features, then models the correlations among parts and propagates local information through GCNs. SGAT [55] designs ISGAT and ESGAT to utilize intrinsic and extrinsic structural relationships, extracting useful information from relevant nodes to update vehicle representations.
3) *Self-Supervised Feature Extraction Methods:* SAVER [39] and SSBVER [41] improve the accuracy of vehicle Re-ID by enhancing image features and utilizing knowledge distillation, respectively.
4) *Other Methods:* MUSP [56] uses multiple attentions to provide part-aware attention weights and robust features. ANet [57] utilizes attribute cues to augment vehicle features, thereby enhancing the model's discriminative power. HRCN [58] construct robust feature representations by integrating region-specific features and cross-level features to complement the original high-level features.

1) Comparison on the VeRi776 Dataset: Table II shows performance comparison of the proposed method and other state-of-the-art approaches on the VeRi776 [9]. Our model outperforms all mentioned methods without using reranking. The proposed PEFN method achieves the highest mAP (85.15%), CMC1 (98.03%), and CMC5 (98.87%). It outperforms the HRCN by 2.05% mAP and 0.73% CMC1, and the HCI-Net 1.35% mAP and 1.43% CMC1. Compared to other methods, our approach more effectively captures both the complete and significant global features and local features, hence demonstrating superior performance.

2) Comparison on the VehicleID Dataset: We compare the PEFN with other methods on VehicleID [10] dataset, and report the CMC1 and CMC5 achieved on three test subsets in Table III. Across all metrics except for the CMC5 score on the small subset where we place second, our method PEFN

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PERFORMANCE (%) COMPARISONS ON VERI776. BOLD AND UNDERLINE ROW REPRESENT THE FIRST AND SECOND, RESPECTIVELY

Method	mAP	CMC1	CMC5
RAM[30]	61.5	88.6	94.0
AAVER[26]	61.18	88.97	94.70
SAN[29]	72.5	93.3	97.1
SGAT[55]	65.66	89.69	-
PVEN[24]	79.5	95.6	98.4
SAVER[39]	79.6	96.4	98.6
PCRNet[54]	78.6	95.4	98.4
ANet[57]	81.2	96.8	98.4
HRCN[58]	83.1	97.3	**98.9**
HPGN[36]	80.18	96.72	-
MUSP[56]	78.0	95.6	97.9
GiT[37]	83.04	96.86	-
SSBVER[41]	82.11	97.08	98.45
HCI-Net[53]	83.8	96.6	**98.9**
PEFN(Ours)	**85.15**	**98.03**	98.87

TABLE III
PERFORMANCE (%) COMPARISONS ON VEHICLEID. BOLD AND UNDERLINE ROW REPRESENT THE FIRST AND SECOND, RESPECTIVELY

Method	Small		Medium		Large	
	CMC1	CMC5	CMC1	CMC5	CMC1	CMC5
RAM[30]	75.2	91.5	72.3	87.0	67.7	84.5
AAVER[26]	74.69	93.82	68.62	89..95	63.54	85.64
SAN[29]	79.7	94.3	78.4	91.3	75.6	88.3
SGAT[55]	78.12	-	73.98	-	71.87	-
PVEN[24]	84.7	97.0	80.6	94.5	77.8	92.0
SAVER[39]	79.9	95.2	77.6	91.1	75.3	88.3
PCRNet[54]	86.6	98.1	82.2	96.3	80.4	94.2
ANet[57]	87.9	97.8	82.8	96.2	80.5	94.6
HRCN[58]	88.2	**98.4**	81.4	96.6	80.2	94.4
HPGN[36]	83.91	-	79.97	-	77.32	-
MUSP[56]	84.5	97.6	82.3	95.1	80.6	93.1
GiT[37]	84.65	-	80.52	-	77.94	-
SSBVER[41]	85.61	97.72	81.62	94.92	78.91	92.60
HCI-Net[53]	83.8	96.5	79.4	92.7	76.4	91.2
PEFN(Ours)	**94.75**	98.12	**92.13**	**96.94**	**90.79**	**96.13**

leads the rankings in both mAP and CMC1 evaluations. The feature supplement techniques employed by our method enable it to quickly and accurately perform Re-ID on this dataset, which is primarily composed of front and rear view images, and significantly outperforms other methods.

E. Ablation Studies and Analysis

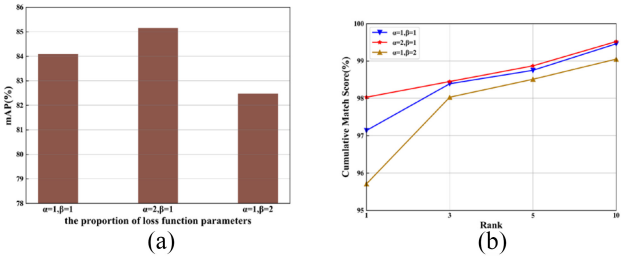
1) Effectiveness of Network Components: We conduct ablation study on VeRi776 dataset to validate the effectiveness of each part component in our model. PEFN w/o BHP indicates the PEFN model without the EnHi branch for multilevel fusion. PEFN w/o IDFR means that the GLAS branch of the PEFN model does not undergoes split, supplementation and aggregate operations. PEFN w/o sup implies the PEFN model removes the residual supplementation(rub). PEFN uses entire architecture as described in Section III. As shown in Table IV, first, we can observe that PEFN achieves better accuracy than others by a large margin. It is because patch supplementation encourages the model to concentrate on the overlooked features and feature fusion enables the model to better comprehend contextual information. Second, the

TABLE IV
ABLATION STUDY ABOUT EACH PART OF PEFN ON VERI776

Settings	mAP	CMC1	CMC5
PEFN w/o BHP	83.13	96.72	98.69
PEFN w/o IDFR	84.47	96.78	98.51
PEFN w/o sup	84.58	96.84	98.45
PEFN	**85.15**	**98.03**	**98.87**

TABLE V
PERFORMANCE (%) COMPARISON AMONG DIFFERENT SPLIT SIZE ON VERI776

Name	Size	mAP	CMC1	CMC5
PEFN1	(3 , 3)	84.46	**98.51**	**99.05**
PEFN2	(6 , 6)	84.57	97.08	98.93
PEFN	(9 , 9)	**85.15**	98.03	98.87



(a) (b)

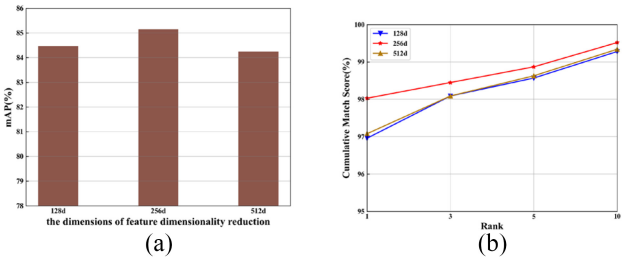
Fig. 6. Parameter analysis of the loss weight α and β on VeRi776. (a) mAP (%) values obtained with different parameters. (b) CMC curves with different parameters.

distinction between PEFN w/o IDFR and PEFN w/o sup is solely in the feature map partitioning and reassembly, yet the results exhibit a variance. It is evident that partition the feature map into some pieces followed attention module enhances the model's proficiency in capturing comprehensive and localized characteristic.

2) Influence of Partition Strategy in IDFR: IDFR is a key component of PEFN, thus influence of split size is evaluated. The input size of IDFR module is (18,18), so there are three splitting sizes: (3, 3), (6, 6), and (9, 9). Dividing according to the mentioned scale results in obtaining pieces in number of 36, 9, and 4 respectively. PEFN, PEFN1 and PEFN2 method respectively employ (9, 9), (3, 3), and (6, 6) splitting size. From Table V, the mAP of PEFN is considerably higher than that of PEFN1 and PEFN2, yet its CMC1 and CMC5 fall behind those of PEFN1. The results imply that appropriate splitting size contributes to promote vehicle Re-ID balanced performance. At the same time, applying attention mechanisms on feature map of different splitting sizes requires careful selection of the convolution kernel size.

3) Analysis of Loss Function Parameters: Our loss function takes the form as shown in (10). To achieve the optimal proportion of the loss function, we conducted tests with three combinations, and the results are shown in Fig. 6. To ensure a consistent and optimal performance, we set $\alpha = 2$ and $\beta = 1$ in all our experiments. The results indicate that an appropriate proportion of cross-entropy loss greatly helps the model to converge more quickly during the training process and enhances the model's discriminative ability for difficult samples.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 7. Parameter analysis of the feature dimension on VeRi-776. (a) mAP values obtained at different dimensions. (b) CMC curves at different dimensions.

TABLE VI
EFFICIENCY METRICS RESULTS OF DIFFERENT MODELS IN THE VEHICLE RE-ID PROCESS

Method	Params (M)	Dims	Inference Time(ms/image)	Memory (MB)
PVEN[24]	59.2	10240	11.79	603
SAVER[39]	31	2048	5.06	178
ANet[57]	67	2048	-	-
HRCN[58]	55.4	3584	10.84	260
HPGN[36]	27.71	256	27.70	-
GiT[37]	35.1	768	54.29	-
SSBVER[41]	23.5	2048	4.55	122
PEFN(Ours)	67	7168	35.7	255.7

4) Analysis of Feature Reduction Parameters: Before the features enter the loss calculation, they need to be dimension-reduced through convolution. For this reason, we further examined the impact of different feature dimensions on the performance of the method. As shown in Fig. 7, different feature dimensions have a significant impact on performance. The method achieved the highest mAP (85.15%) and CMC1 (98.03%) when using a feature dimension of 256. However, when using feature dimensions of 128 and 512, there was a decline in both metrics to varying degrees.

5) Analysis of Computational Overhead: As shown in Table VI, some results regarding computational overhead are presented. Based on the evaluation results in Tables II and III, we select HRCN and GiT as comparison models. The embedding vectors of PFEN are twice that of HRCN; the inference speed of PEFN is slightly inferior to HRCN, and the memory consumption is slightly less than HRCN. Moreover, when dealing with embedding vectors several times larger than GiT, PEFN has a better inference speed than it. The inference time is related to the complexity of the model; the longer the feature extraction process, the slower the inference time.

6) Sorting Visualization: To visualize our proposed PEFN method's effectiveness, the rank-10 retrieval results of five different vehicles query images from VeRi776 are shown in Fig. 8. The first image in each row represents query image. Images with green boxes and red boxes are correct and incorrect retrieval results, respectively. As shown in Table II, the results in Fig. 8 reflect the strong retrieval capability of the proposed PEFN model given the query images from the designated viewpoint. It is noteworthy that the second retrieved image and the fifth retrieved image have the same occlusion and exceedingly similar vehicle shapes, yet PEFN still exhibits strong robust retrieval performance.

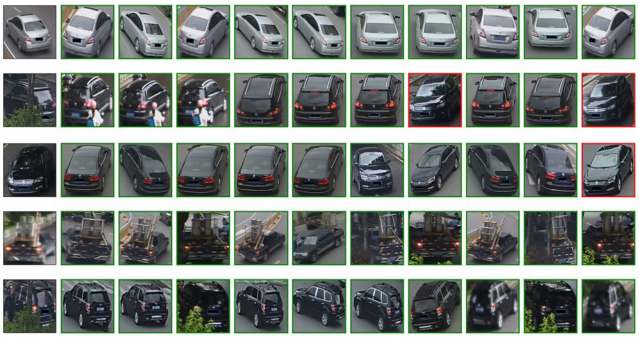
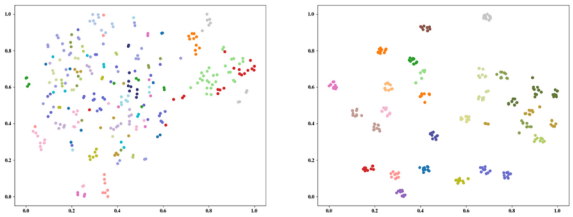


Fig. 8. Rank-10 visualization examples on VeRi776. The red and green boxes indicate that the image is from a different vehicle and the same vehicle as the query image, respectively.



(a) (b)

Fig. 9. t-SNE visualization of ResNet-50 and PEFN for 300 samples from 30 classes. (a) Visualization result of ResNet-50. (b) Visualization result of PEFN.

7) Discriminability of Feature Representations: We randomly selected 300 samples from 30 classes in the VeRi-776 dataset and visualized the output features of the pretrained backbone network ResNet-50 and PEFN using t-SNE, respectively. The visualization results use 30 different colors to correspond to different classes, with features of images from the same class having the same color. Fig. 9(a) shows the visualization result of ResNet-50, which has a confused ability to distinguish vehicle classes. Fig. 9(b) shows that the PEFN model can more compactly cluster features of vehicles from the same class and separate features of vehicles from different classes farther apart. This indicates that PEFN has a stronger ability to distinguish feature representations and can effectively address the problem of high interclass similarity.

F. Industrial Application

We state that the PEFN will bring advancements in several key areas of industrial applications, including intelligent transportation, automotive safety, and surveillance systems [49], [59], [60].

1) Intelligent Transportation: The PEFN framework can be seamlessly integrated into IoT-enabled intelligent transportation systems, particularly for real-time vehicle Re-ID and tracking. For instance, PEFN can be deployed on edge devices, such as IoT-enabled cameras and sensors, to enhance the accuracy of vehicle identification under varying lighting conditions. The lightweight design of PEFN makes it suitable for deployment on resource-constrained edge devices, enabling rapid local data processing, reduced latency, and minimized bandwidth usage while ensuring real-time performance. This

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

capability is crucial for applications requiring immediate responses, such as traffic violation detection and smart parking management.

2) Automotive Safety: In the realm of autonomous driving and connected vehicles, PEFN can significantly enhance the visual perception systems of vehicles, particularly in challenging scenarios, such as diverse weather conditions, lighting changes, and complex traffic environments. By integrating PEFN with IoT-enabled systems, vehicles can share real-time visual data with other connected devices and infrastructure (e.g., traffic lights, road sensors, and cloud platforms), enabling collaborative decision-making and improving overall road safety. For instance, PEFN can process and analyze data from onboard cameras and IoT sensors to detect potential hazards, providing drivers or autonomous systems with timely warnings.

3) Surveillance Systems: Given that PEFN have enhanced the accuracy of vehicle identification, it can also achieve efficient vehicle tracking and violation detection (such as red-light running and speeding) in video surveillance systems. Furthermore, Logistics and Supply Chain Management is one of the industrial application scenarios for PEFN. By integrating PEFN with IoT-enabled systems, logistics vehicles can be equipped with GPS and IoT sensors to track their real-time location, status, and environmental conditions. This data can be transmitted to a centralized cloud platform, where PEFN processes and analyzes it to optimize route planning, reduce delivery times, and minimize fuel consumption.

V. Conclusion and Future Work

This article introduces the PEFN, a novel method designed to improve the extraction of both global and discriminative local features from vehicle images. The core components of the network include the IDFR module and the EnHi branch. The IDFR module enhances feature maps that have already passed through the attention mechanism, applying small patches to further refine the features, thereby enriching both global and local information. By reducing the emphasis on intrainstance unique information, the module enables a stronger focus on invariant and salient features, which is essential for accurate vehicle Re-ID. The EnHi branch facilitates the fusion of information from feature maps at different depths, allowing the model to better capture contextual relationships and improve the model's understanding of vehicle features in various contexts. Extensive experimental results validate the effectiveness of our approach. Specifically, our model achieves an impressive 85.15% mAP on the VeRi776 dataset, outperforming many state-of-the-art methods in terms of accuracy and robustness. Our method significantly addresses the challenges of interclass similarity and intraclass variance, which are common issues in vehicle Re-ID tasks, providing a promising solution for this problem.

However, despite the success of our proposed method, there remains considerable room for further improvement. Current vehicle Re-ID models primarily focus on image-based features, which could limit their ability to capture all relevant information for accurate identification. In the future, incorporating additional modalities, such as GPS data, timestamps, or even sensor-based data from vehicles, could offer valuable complementary information, enhancing the overall performance. GPS data and timestamps, as spatiotemporal data, provide vehicle locations and trajectories, which can logically filter out incorrect matches quickly. However, this requires a high-precision synchronization mechanism. Textual descriptions of vehicle information offer additional criteria for discrimination when visual features are insufficient. Integrating these multimodal data necessitates designing complex fusion mechanisms that balance feature extraction and interaction across different modalities. Moreover, the annotation and acquisition of multimodal data are costly, making the integration of multimodal data quite challenging. Nevertheless, exploring multimodal learning techniques that combine these diverse data sources presents an important direction for future research. Furthermore, improving the model's efficiency in real-time applications and extending its applicability to more complex and dynamic environments are key areas for further development.

References

[1] J. M. Guo, C. H. Hsia, K. Wong, J. Y. Wu, Y. T. Wu, and N. J. Wang, "Nighttime vehicle lamp detection and tracking with adaptive mask training," *IEEE Trans. Veh. Technol.*, vol. 65, pp. 4023–4032, Jun. 2016.

[2] Y. Z. Zhao, C. H. Shen, H. B. Wang, and S. Y. Chen, "Structural analysis of attributes for vehicle re-identification and retrieval," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 2, pp. 723–734, Feb. 2020.

[3] S. V. Huynh, N. H. Nguyen, N. T. Nguyen, V. T. Q. Nguyen, C. Huynh, and C. Nguyen, "A strong baseline for vehicle re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 4142–4149.

[4] H. Chen, B. Lagadec, and F. Bremond, "Partition and reunion: A two-branch neural network for vehicle re-identification," in *Proc. CVPR Workshops*, 2019, pp. 184–192.

[5] X. C. Liu, W. Liu, T. Mei, and H. D. Ma, "PROVID: Progressive and multimodal vehicle reidentification for large-scale urban surveillance," *IEEE Trans. Multimedia*, vol. 20, no. 3, pp. 645–658, Mar. 2018.

[6] X. B. Liu, S. L. Zhang, X. Y. Wang, R. C. Hong, and Q. Tian, "Group-group loss-based global-regional feature learning for vehicle re-identification," *IEEE Trans. Image Process.*, vol. 29, pp. 2638–2652, 2020.

[7] H. Sheng et al., "Combining pose invariant and discriminative features for vehicle reidentification," *IEEE Internet Things J.*, vol. 8, no. 5, pp. 3189–3200, Mar. 2021.

[8] F. Shen et al., "An efficient multiresolution network for vehicle reidentification," *IEEE Internet Things J.*, vol. 9, pp. 9049–9059, 2021.

[9] X. C. Liu, W. Liu, T. Mei, and H. D. Ma, "A deep learning-based approach to progressive vehicle re-identification for urban surveillance," in *Proc. 14th Eur. Conf. Comput. Vis. (ECCV)*, Amsterdam, The Netherlands, 2016, pp. 869–884.

[10] H. Y. Liu, Y. H. Tian, Y. W. Wang, L. Pang, and T. J. Huang, "Deep relative distance learning: Tell the difference between similar vehicles," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, 2016, pp. 2167–2175.

[11] Y. H. Lou, Y. Bai, J. Liu, S. Q. Wang, L. Y. Duan, and I. C. Soc, "VERI-wild: A large dataset and a new method for vehicle re-identification in the wild," in *Proc. 32nd IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, 2019, pp. 3230–3238.

[12] H. Y. Guo, C. Y. Zhao, Z. W. Liu, J. Q. Wang, and H. Q. Lu, "Learning coarse-to-fine structured feature embedding for vehicle re-identification," in *Proc. 32nd AAAI Conf. Artif. Intell. 30th Innovat. Appl. Artif. Intell. Conf. 8th AAAI Symp. Educ. Adv. Artif. Intell.*, New Orleans, LA, USA, 2018, pp. 6853–6860.

[13] K. Yan, Y. H. Tian, Y. W. Wang, W. Zeng, and T. J. Huang, "Exploiting multi-grain ranking constraints for precisely searching visually-similar vehicles," in *Proc. 16th IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, 2017, pp. 562–570.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

[14] W. Zeng, T. Wang, J. Cao, J. Wang, and H. Zeng, "Clustering-guided pairwise metric triplet loss for person reidentification," *IEEE Internet Things J.*, vol. 9, no. 16, pp. 15150–15160, Aug. 2022.

[15] X. Liu, Z. Zhou, C. Niu, and Q. Wu, "Visual–textual alignment for generalizable person reidentification in Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 15, pp. 13865–13875, Aug. 2023.

[16] F. Yang, W. Li, B. Liang, and J. Zhang, "Spatiotemporal interaction transformer network for video-based person reidentification in Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12537–12547, Jul. 2023.

[17] J. Q. Zhu, H. Q. Zeng, Z. Lei, S. C. Liao, L. X. Zheng, and C. H. Cai, "A shortly and densely connected convolutional neural network for vehicle re-identification," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Beijing, China, 2018, pp. 3285–3290.

[18] Z. D. Zheng, T. Ruan, Y. C. Wei, Y. Yang, and T. Mei, "VehicleNet: Learning robust visual representation for vehicle re-identification," *IEEE Trans. Multimedia*, vol. 23, pp. 2683–2693, 2021.

[19] R. H. Chu, Y. F. Sun, Y. D. Li, Z. Liu, C. Zhang, and Y. C. Wei, "Vehicle re-identification with viewpoint-aware metric learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, 2019, pp. 8281–8290.

[20] Y. Zhang, D. Liu, and Z.-J. Zha, "Improving triplet-wise training of convolutional neural network for vehicle re-identification," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, 2017, pp. 1386–1391.

[21] R. Kumar, E. Weill, F. Aghdasi, and P. Sriram, "Vehicle re-identification: An efficient baseline using triplet embedding," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Budapest, Hungary, 2019, pp. 1–9.

[22] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Boston, MA, USA, 2015, pp. 1–9.

[23] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[24] D. C. Meng et al., "Parsing-based view-aware embedding network for vehicle re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 7101–7110.

[25] X. B. Chen, H. Z. Yu, F. Zhao, Y. Hu, and Z. Y. Li, "Global–local discriminative representation learning network for viewpoint-aware vehicle re-identification in intelligent transportation," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, Jul. 2023.

[26] P. Khorramshahi, A. Kumar, N. Peri, S. S. Rambhatla, J. C. Chen, and R. Chellappa, "A dual-path model with adaptive attention for vehicle re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, 2019, pp. 6131–6140.

[27] T.-S. Chen, M.-Y. Lee, C.-T. Liu, and S.-Y. Chien, "Viewpoint-aware channel-wise attentive network for vehicle re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 2448–2455.

[28] W. Zhu et al., "A dual self-attention mechanism for vehicle re-identification," *Pattern Recognit.*, vol. 137, May 2023, Art. no. 109258.

[29] J. J. Qian, W. Jiang, H. Luo, and H. Y. Yu, "Stripe-based and attribute-aware network: A two-branch deep model for vehicle re-identification," *Meas. Sci. Technol.*, vol. 31, Sep. 2020, Art. no. 95401.

[30] X. Liu, S. Zhang, Q. Huang, and W. Gao, "Ram: A region-aware deep model for vehicle re-identification," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, 2018, pp. 1–6.

[31] X. Ma, K. Zhu, H. Guo, J. Wang, M. Huang, and Q. Miao, "Vehicle re-identification with refined part model," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, 2019, pp. 603–606.

[32] X. Pei, X. Deng, N. N. Xiong, S. Mumtaz, and J. Wu, "Complex graph analysis and representation learning: Problems, techniques, and applications," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 5, pp. 4990–5007, Sep./Oct. 2024.

[33] C. Wang, H. Fu, and H. Ma, "Global structure graph guided fine-grained vehicle recognition," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2020, pp. 1913–1917.

[34] A. M. N. Taufique and A. Savakis, "LABNet: Local graph aggregation network with class balanced loss for vehicle re-identification," *Neurocomputing*, vol. 463, pp. 122–132, Nov. 2021.

[35] R. Zhang, X. Zhong, X. Wang, W. Huang, W. Liu, and S. Zhao, "Graph-based structural attributes for vehicle re-identification," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, 2022, pp. 1–6.

[36] F. Shen, J. Q. Zhu, X. B. Zhu, Y. Xie, and J. C. Huang, "Exploring spatial significance via hybrid pyramidal graph network for vehicle re-identification," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 8793–8804, Jul. 2022.

[37] F. Shen, Y. Xie, J. Q. Zhu, X. B. Zhu, and H. Q. Zeng, "GiT: Graph interactive transformer for vehicle re-identification," *IEEE Trans. Image Process.*, vol. 32, pp. 1039–1051, 2023.

[38] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Annu. Conf. Neural Inf. Process. Syst. (NIPS)*, Long Beach, CA, USA, 2017, pp. 1–11.

[39] P. Khorramshahi, N. Peri, J.-c. Chen, and R. Chellappa, "The devil is in the details: Self-supervised attention for vehicle re-identification," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K., 2020, pp. 369–386.

[40] M. Li, X. M. Huang, and Z. M. Zhang, "Self-supervised geometric features discovery via interpretable attention for vehicle re-identification and beyond," in *Proc. 18th IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 194–204.

[41] P. Khorramshahi, V. Shenoy, and R. Chellappa, "Robust and scalable vehicle re-identification via self-supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 5295–5304.

[42] R. Zhou et al., "Self-attention mechanism enhanced user interests modeling for personalized recommendation services in cyber-physical-social systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 6, pp. 3835–3846, Nov./Dec. 2021.

[43] X. Zhang, H. Cheng, Z. Yu, and N. N. Xiong, "Design and analysis of an efficient multiresource allocation system for cooperative computing in Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 16, pp. 14463–14477, Aug. 2022.

[44] K. M. He, X. Y. Zhang, S. Q. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, 2016, pp. 770–778.

[45] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE-Comput.-Soc. Conf. Comput. Vis. Pattern Recognit. Workshops*, Miami Beach, FL, USA, 2009, pp. 248–255.

[46] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, 2016, pp. 2818–2826.

[47] A. Hermans, L. Beyer, and B. Leibe, "In defense of the triplet loss for person re-identification," 2017, *arXiv:1703.07737*.

[48] S. H. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. 15th Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, 2018, pp. 3–19.

[49] X. Liu et al., "Secure computing protocols for Internet of Things data fusion based on set intersection and union against malicious enemies," *IEEE Internet Things J.*, vol. 11, no. 14, pp. 24918–24932, Jul. 2024.

[50] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, Munich, Germany, 2015, pp. 234–241.

[51] T. Y. Lin, A. RoyChowdhury, and S. Maji, "Bilinear CNN models for fine-grained visual recognition," in *Proc. IEEE Int. Conf. Comput. Vis.*, Santiago, Chile, 2015, pp. 1449–1457.

[52] L. Zheng, L. Y. Shen, L. Tian, S. J. Wang, J. D. Wang, and Q. Tian, "Scalable person re-identification: A benchmark," in *Proc. IEEE Int. Conf. Comput. Vis.*, Santiago, Chile, 2015, pp. 1116–1124.

[53] K. Sun et al., "Heterogeneous context interaction network for vehicle re-identification," *Neural Netw.*, vol. 169, pp. 293–306, Jan. 2024.

[54] X. Liu, W. Liu, J. Zheng, C. Yan, and T. Mei, "Beyond the parts: Learning multi-view cross-part correlation for vehicle re-identification," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 907–915.

[55] Y. Zhu, Z.-J. Zha, T. Zhang, J. Liu, and J. Luo, "A structured graph attention network for vehicle re-identification," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 646–654.

[56] S. Lee, T. Woo, and S. H. Lee, "Multiple soft attention network for vehicle re-identification," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Macau, China, 2022, pp. 2903–2907.

[57] R. Quispe, C. L. Lan, W. J. Zeng, and H. Pedrini, "AttributeNet: Attribute enhanced vehicle re-identification," *Neurocomputing*, vol. 465, pp. 84–92, Nov. 2021.

[58] J. Zhao, Y. Zhao, J. Li, K. Yan, and Y. Tian, "Heterogeneous relational complement for vehicle re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 205–214.

[59] X. Deng, J. Yin, P. Guan, N. N. Xiong, L. Zhang, and S. Mumtaz, "Intelligent delay-aware partial computing task offloading for multiuser Industrial Internet of Things through edge computing," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 2954–2966, Feb. 2023.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.

[60] J. Zhang, J. Ma, Y. Yang, X. Liu, and N. N. Xiong, "Revocable and privacy-preserving decentralized data sharing framework for fog-assisted Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 10446–10463, Jul. 2022.



Wenying He received the Ph.D. degree from Tianjin University, Tianjin, China, in 2021.

She is currently a Lecturer with the School of Artificial Intelligence, Hebei University of Technology, Tianjin. Her research interests include machine learning, computer vision, and bioinformatics.



Feiyu Wang is currently pursuing the master's degree with the School of Artificial Intelligence and Data Science, Hebei University of Technology, Tianjin, China.

His research interests include computer vision and vehicle reidentification.



Yude Bai received the Ph.D. degree from Tianjin University, Tianjin, China, in 2023.

He is a Lecturer with Tiangong University, Tianjin, China. His research focuses on deep learning, malware detection, and software vulnerability analysis.



Neal N. Xiong (Senior Member, IEEE) received the first Ph.D. degree in sensor system engineering from Wuhan University, Wuhan, China, in 2007, and the second Ph.D. degree in dependable communication networks from Japan Advanced Institute of Science and Technology, Nomi, Japan, in 2008.

He is currently an Associate Professor (5th year) with the Department of Mathematics and Computer Science, Northeastern State University, Tahlequah, OK, USA. His research interests include cloud computing, security and dependability, parallel and distributed computing, networks, and optimization theory.



Guangquan Xu (Member, IEEE) received the Ph.D. degree from Tianjin University, Tianjin, China, in March 2008.

He is a Full Professor with Tianjin Key Laboratory of Advanced Networking (TANK), College of Intelligence and Computing, Tianjin University. He is also an Adjunct Professor with Qingdao Huanghai University, Qingdao, China. His research interests include cyber security and trust management.

Prof. Xu is a member of CCF.



Fei Guo received the B.S. and Ph.D. degrees from Shandong University, Jinan, China, in 2007 and 2012, respectively.

She was a Postdoctoral Fellow with the City University of Hong Kong, Hong Kong, from 2012 to 2013. She is currently a Professor with the School of Computer Science and Engineering, Central South University, Changsha, China.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 11:00:39 UTC from IEEE Xplore. Restrictions apply.